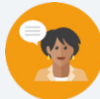


## 9.8 Surface Area of Three-Dimensional Figures

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                      |                                                                                                                                                                                                                                                                                           |                                                                                           |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|  Benchmark Focus     | MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.                                                                                                                                                                                                                                                             |  Learning Targets | <ul style="list-style-type: none"><li>I can solve mathematical problems involving the surface area of cylinders, pyramids, prisms, cones, and spheres.</li><li>I can solve mathematical problems involving the lateral area of cylinders, pyramids, prisms, cones, and spheres.</li></ul> |                                                                                           |
|  Benchmark Coherence | Building On<br>MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.                                                                                                                                                                                                                                                                                        |                                                                                                      | Working Toward<br>N/A                                                                                                                                                                                                                                                                     |                                                                                           |
|  Essential Questions | <ul style="list-style-type: none"><li>How can I solve mathematical problems involving the surface area of cylinders, pyramids, prisms, cones, and spheres?</li><li>How can I solve mathematical problems involving the lateral area of cylinders, pyramids, prisms, cones, and spheres?</li><li>How can I derive the formulas for lateral area and surface area of a cylinder, pyramid, prism, cone, and sphere?</li></ul>               |                                                                                                      |                                                                                                                                                                                                                                                                                           |                                                                                           |
|  Lesson Narrative    | This lesson guides students through the process of deriving formulas for lateral area and surface area of cylinders, prisms, and pyramids. The cone and sphere formulas are given. Students then work through a series of mathematical problems relating to finding the lateral areas and surface areas of various three-dimensional shapes. This lesson does not address real-world problems, as these will be addressed in Lesson 9.9. |                                                                                                      |                                                                                                                                                                                                                                                                                           |                                                                                           |
|  Component Summary | 9.8.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                  | Choosing which image doesn't belong ( <i>SE p. 94</i> )                                              | 9.8.5 Guided Instruction (5-10 min.)                                                                                                                                                                                                                                                      | Finding lateral areas and surface areas of figures ( <i>SE p. 97</i> )                    |
|                                                                                                       | 9.8.2 Collaboration (5 min.)                                                                                                                                                                                                                                                                                                                                                                                                             | Determining lateral area and surface area ( <i>SE p. 94</i> )                                        | 9.8.6 Guided Practice (5 min.)                                                                                                                                                                                                                                                            | Finding lateral area and surface area of a pyramid, prism, and sphere ( <i>SE p. 98</i> ) |
|                                                                                                       | 9.8.3 Activity (10 min.)                                                                                                                                                                                                                                                                                                                                                                                                                 | Exploring nets to determine lateral area and surface area ( <i>SE p. 95</i> )                        | 9.8.7 Your Turn! (5 min.)                                                                                                                                                                                                                                                                 | Finding lateral area and surface area of a cone and cylinder ( <i>SE p. 99</i> )          |
|                                                                                                       | 9.8.4 Bring It Together (5 min.)                                                                                                                                                                                                                                                                                                                                                                                                         | Formalizing the formulas for lateral area and surface area ( <i>SE p. 96</i> )                       | 9.8.8 Wrap-Up (~5 min)                                                                                                                                                                                                                                                                    | Analyzing work finding surface area of a pyramid ( <i>SE p. 99</i> )                      |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                      | Materials                                                                                                                                                                                                                                                                                 |                                                                                           |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Lateral area</li><li>Surface area</li></ul><br><u>Additional Terms:</u> N/A                                                                                                                                                                                                                                                                                                      |                                                                                                      | <ul style="list-style-type: none"><li>Nets for Gallery Walk</li><li>(Optional) Colored Pencils, Markers, Pens</li><li>(Optional) Various 3D Models</li></ul>                                                                                                                              |                                                                                           |
|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                      | Additional Preparation <ul style="list-style-type: none"><li>This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (i.e. Word Wall, math cards, etc.).</li></ul>                                                     |                                                                                           |

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips              | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may use the incorrect formulas when finding surface area.</li><li>Students may use the surface area formula to find lateral area instead of the part of the formula that uses lateral area.</li><li>Students may not remember to divide the diameter by two when finding radius.</li><li>Students may calculate perimeter instead of area.</li><li>Students may not round final answers to the correct place value.</li></ul> | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Consider paring down parts of the 9.8.5 Guided Instruction, 9.8.6 Guided Practice, or 9.8.7 Your Turn! in the event of a time crunch by allowing student choice of individual questions.</li><li>Consider having three-dimensional models and/or nets as another visual for students.</li></ul>                                                                                |
|                                                                                                                | <b>Literacy and Language Routines</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Mathematical Thinking and Reasoning (MTR) Standard</b>                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                                | <b>Which One Doesn't Belong?</b><br>9.8.1 Warm-Up in this lesson is a Which One Doesn't Belong. Allow students to share their specific reasonings as to which one doesn't belong in the set. Pay specific attention to students' vocabulary here and create an environment that is no risk and that is accessible to all students.                                                                                                                                                                        | <b>MA.K12.MTR.4.1: Discussion and Reflection</b><br>This lesson has multiple opportunities for student discussion built in and structured. 9.8.2 Collaboration requires that students discuss to develop the formulas for surface area, and the 9.8.4 Bring It Together has students discussing the various formulas. MA.K12.MTR.4.1 also has students discuss and compare the efficiency of methods, which is also present within the lesson. |
| <br>Differentiate Instruction | 9.8.2 Collaboration and 9.8.3 Activity are appropriate to differentiate instruction for students. As students derive formulas, some may need more scaffolding. Provide digital visuals using technology. Consider using GeoGebra to provide the visuals for students.                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                |

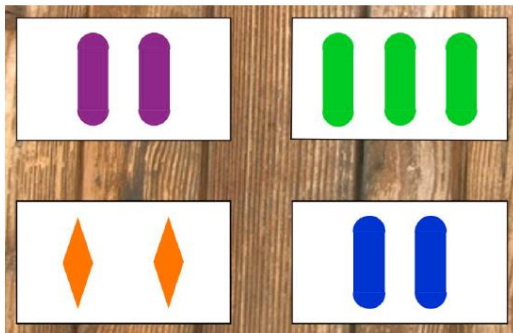
## 9.8.1 Warm-Up: Which One Doesn't Belong?

**Component Overview:** Students begin a low-floor, high-ceiling task by stating which card does not belong in the set and why.



### 9.8.1 Warm-Up: Which One Doesn't Belong?

- Which one doesn't belong? Explain your answer.



Answers will vary.

The image with the 2 orange rhombi does not belong because the rhombi are not composite figures.

The image with the purple figures does not belong because purple is not a primary color.

The image with the green figures contains 3 composite figures.

The image with the 2 blue figures because in the second row, they are not rhombi.



Allow students to think about which card in the set does not belong. Provide an opportunity for several students to share their responses.



For a kinesthetic approach to 9.8.1 Warm-Up, have students stand in separate corners of the room and hear a response or two from each group. Students will stand in the corner that matches their response.

A - top left (purple card)

B - top right (green card)

C - bottom left (orange card)

D - bottom right (blue card)

## 9.8.2 Collaboration: Using Nets to Find Surface Area

**Component Overview:** Students will begin working with a partner to determine formulas for lateral area and surface area of various three-dimensional figures. Assign a three-dimensional figure to each group and give time for students to derive the formula using the prompts in the questions.



### 9.8.2 Collaboration: Using Nets to Find Surface Area



The **lateral area** of a three-dimensional figure is the sum of the area of the figure's faces.



The **surface area** of a three-dimensional figure is the sum of the lateral area of the figure's faces and the area of the figure's bases.

1. Work with your partner to determine a formula for the lateral area and the surface area of the figure you have been given. Complete the table.

|                      |  |  |  |  |
|----------------------|--|--|--|--|
| Our figure was:      |  |  |  |  |
| <b>Dimension</b>     |  |  |  |  |
| <b>Variable Used</b> |  |  |  |  |

*Answers will vary. See the table in the Gallery Walk below.*



*Students will not be given specific numbers to use, so only variables will be used here.*

*Teachers may need to review the difference between the faces and bases of a figure.*



*Teachers assign each pair one of the six figures from the 9.8.3 Activity to determine the surface area. Each pair can be given a larger version of the figures or use the figures in the table. Once each pair has completed 9.8.2 Collaboration, then place students in groups based on their figure to create information for the Gallery Walk.*

*Teachers may need to explain why the student must define the variables they used.*

## 9.8.3 Activity: Gallery Walk

**Component Overview:** Students are recording the formulas from the other groups onto their individual papers. Post the students' work around the classroom and have students complete a Gallery Walk. Give students no more than one to two minutes per station. Create a structure that allows students to move through this as efficiently as possible.



### 9.8.3 Activity: Gallery Walk

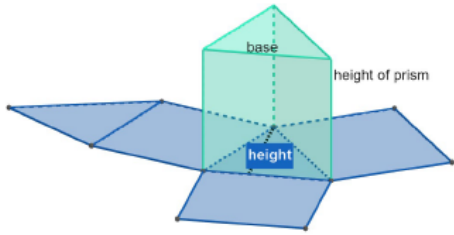
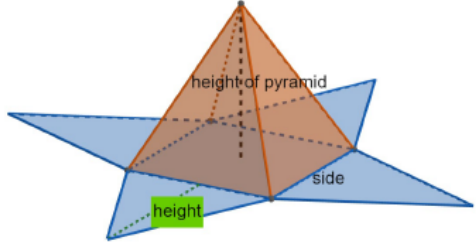
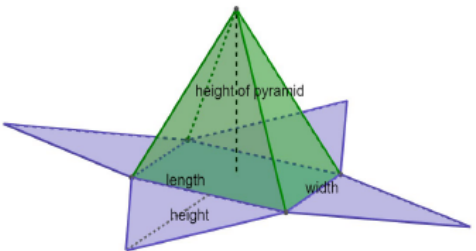
- During the gallery walk, record the formulas. Make any useful notes on each figure's net.

| Figure                                                                                                                                                                                         | Net |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Cylinder</b><br><br>Lateral Area = $2\pi rh$ ,<br><i>where <math>r</math> = radius and <math>h</math> = height.</i><br>Surface Area = $2(\pi r^2 + \pi rh)$                                 |     |
| <b>Cube</b><br><br>Lateral Area = $4s^2$ ,<br><i>where <math>s</math> = side.</i><br>Surface Area = $6s^2$                                                                                     |     |
| <b>Rectangular Prism</b><br><br>Lateral Area = $2(lh + wh)$ ,<br><i>where <math>l</math> = length, <math>h</math> = height and <math>w</math> = width.</i><br>Surface Area = $2(lh + wh + lw)$ |     |



Students may find during the Gallery Walk that different formulas were created. This will provide a chance for a rich conversation for the teacher to have with the class. Formulas are meant to make finding something easier. Students should not be forced to memorize a formula.

### 9.8.3 Activity: Gallery Walk (continued)

|                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Triangular Prism</b></p> <p>Lateral Area = <math>3bh_p</math>,<br/>where <math>b</math> = base<br/>and <math>h_p</math> = height<br/>of prism.</p> <p>Surface Area = <math>3bh_p + bh</math>, where<br/><math>h</math> = height of<br/>triangular base.</p> |                                                                                                                   |
| <p><b>Square Pyramid</b></p> <p>Lateral Area = <math>2sh_s</math>,<br/>where <math>s</math> = side.</p> <p>Surface Area = <math>2sh_s + s^2</math></p>                                                                                                            |  <p>The height of the triangle that is a face of a pyramid is also known as slant height, <math>h_s</math>.</p>   |
| <p><b>Rectangular Pyramid</b></p> <p>Lateral Area = <math>h_s(l + w)</math>, where<br/><math>l</math> = length and<br/><math>w</math> = width.</p> <p>Surface Area = <math>h_s(l + w) + lw</math></p>                                                             |  <p>The height of the triangle that is a face of a pyramid is also known as slant height, <math>h_s</math>.</p> |



How does the formula for the surface area triangular prism differ from the formula for the surface area of a rectangular prism?



How does the formula for the square pyramid prism differ from the formula for the rectangular pyramid?

## 9.8.4 Bring It Together: Formulas

**Component Overview:** Students are discussing the similarities and differences between the formulas that were derived in 9.8.3 Activity when compared to the formulas given in the chart. Students are also introduced to the formulas for surface area of a sphere and a cone for the first time in this lesson.



### 9.8.4 Bring It Together: Formulas

Using the net of three-dimensional objects can help to find the lateral area and surface area. The net of a cone and a sphere make deriving the formulas for surface area from the net more challenging. A chart of formulas is given, where  $SA$  represents surface area.

| Figure                                                         | Surface Area                           | Variables                                                          |
|----------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------|
| Prism                                                          | $SA = 2B + Ph$                         | $B$ = base, $P$ = perimeter, $h$ = height                          |
| Cylinder                                                       | $SA = 2B + Ch$<br>Or<br>$SA = 2B + Ph$ | $B$ = base, $C$ = circumference (or $P$ = perimeter, $h$ = height) |
| Cone                                                           | $SA = B + \pi r h_s$                   | $B$ = base, $r$ = radius, $h_s$ = slant height                     |
| Regular Pyramid<br>(a pyramid whose base is a regular polygon) | $SA = B + \frac{1}{2}Ph_s$             | $B$ = base, $P$ = perimeter, $h_s$ = slant height                  |
| Sphere                                                         | $SA = 4\pi r^2$                        | $r$ = radius                                                       |

- The formulas in the table may not match the formulas recorded during the gallery walk. Which figures have formulas that are different?

*Answers will vary. All the formulas are different because the formulas in the gallery walk were written in terms of the actual area of each figure's base versus  $B$ , the letter used to represent area of the base of any figure. In addition, the formulas in the gallery walk were not written in terms of  $P$ , the perimeter of the base.*



*This component provides an opportunity for students to discuss how the formulas differ. Provide students time to discuss the similarities and differences in the formulas.*



*Why does a sphere not have a lateral area circled? (There is no specific lateral area for a sphere.)*

## 9.8.4 Bring It Together: Formulas (continued)

2. There are many ways to write and to understand formulas. Discuss with your partner which formulas make more sense to you. Discuss how the formulas in the formula chart can be used to help you remember how to find surface area. Summarize your discussion.  
*Answers will vary. The surface area of a prism,  $SA = 2B + Ph$  makes more sense because it appears to be more direct and simpler to work with.*
3. The chart does not have a formula for lateral area. For each of the figures, except for spheres, circle the part of the surface area formula that gives the lateral area.  
*See the table above.*



*How can you tell which part of the formula is the lateral area? (The part that does not relate to the base of the figure.)*



*Do not spend too much time on having students circle the lateral area, as this will take away from the discussion.*



## 9.8.5 Guided Instruction: Lateral Area and Surface Area

**Component Overview:** Students work through questions finding the lateral area and the surface area with teacher support. This component provides a mixture of questions for students to work through with support from the teacher.



### 9.8.5 Guided Instruction: Lateral Area and Surface Area

1. A square pyramid has a slant height of 7 inches, and the area of the base is 16 square inches. What is the surface area, in square inches?

$$SA = B + \frac{1}{2}Ph_s$$

$$SA = 16 + \frac{1}{2}(16)(7)$$

$$SA = 16 + (8)(7)$$

$$SA = 16 + 56$$

$$SA = 72 \text{ in}^2$$

2. The surface area of a sphere is  $144\pi$  feet. What is the radius of the sphere?

$$SA = 4\pi r^2$$

$$4\pi r^2 = 144\pi$$

$$r^2 = 36$$

$$r = 6 \text{ ft}$$

3. If a cylinder has a height of 16 centimeters and a radius of 8 centimeters.

- a. Find the lateral area. Leave your answer in  $\pi$  form.

$$LA = Ch$$

$$LA = 2\pi(8)(16)$$

$$LA = 256\pi \text{ cm}^2$$

- b. Find the surface area. Leave your answer in  $\pi$  form.

$$SA = 2B + Ch$$

$$SA = 2(\pi 8^2) + 2\pi(8)(16)$$

$$SA = 128\pi + 256\pi$$

$$SA = 384\pi \text{ cm}^2$$



#### Common Misconception

Students may mistakenly find the area instead of the perimeter. When necessary, review how to calculate the perimeter of a figure.



How does question 2 differ from question 1? (Surface area is given, and students must find the radius.)



What does it mean to leave your answers in terms of  $\pi$ ? (Do not multiply by the decimal representation of  $\pi$  when simplifying the formula.)



On question 3b, have students streamline the process by substituting the lateral area into the surface area formula.

## 9.8.5 Guided Instruction: Lateral Area and Surface Area (continued)

4. A right triangular prism has base side lengths of 8, 15, and 17 feet (ft), where 17 ft is the hypotenuse of the triangle. The prism has a height of 7 ft. What is the surface area, in square feet, of the triangular prism?

$$SA = 2B + Ph$$

$$SA = 2\left(\frac{1}{2} \cdot 8 \cdot 15\right) + (40)(7)$$

$$SA = 120 + 280$$

$$SA = 400 \text{ ft}^2$$



*On question 4, encourage students to draw and label an image of a triangular prism prior to finding the surface area.*

## 9.8.6 Guided Practice: Lateral Area and Surface Area

**Component Overview:** Students are working through other mathematical questions relating to finding lateral and surface area.



### 9.8.6 Guided Practice: Lateral Area and Surface Area

1. A pyramid has a square base whose area is 49 square feet with a height of 11.5 feet. Find the lateral area, in square feet, of the pyramid.

$$LA = \frac{1}{2}Ph_s$$

$$LA = \frac{1}{2}(28)(\sqrt{144.5})$$

$$LA = 168.291 \text{ ft}^2$$

2. A rectangular prism has a length of 12.5 inches (in), a width of 8 in., and a height of 5 in. Determine the surface area, in square inches, of the rectangular prism.

$$SA = 2B + Ph$$

$$SA = 2(8)(12.5) + (41)(5)$$

$$SA = 405 \text{ in}^2$$

3. The radius of a sphere is 18.7 feet. Find, in square feet, the surface area of the sphere. Round your answer to thousandths place.

$$SA = 4\pi r^2$$

$$SA = 4\pi(18.7)^2$$

$$SA = 4394.334 \text{ ft}^2$$



To provide students a choice, allow students to use either the formulas that were derived in 9.8.3 Activity or have students use the formulas that were given in the chart.



Students are required to find perimeter if using the formulas from the chart. Students must remember that finding the perimeter requires adding all the sides of the base.

## 9.8.7 Your Turn!

**Component Overview:** Students work independently through two questions related to finding the lateral area of a cone and the surface area of a cylinder.



### 9.8.7 Your Turn!

- Find the lateral area, in square meters, of a cone that has a radius of 3 meters (m) and a slant height of 8.5 m. Round your answer to thousandths place.

$$LA = \pi r h_s$$

$$LA = \pi(3)(8.5)$$

$$LA = 80.111 m^2$$

- A cylinder has a radius of 5 inches (in) and a height of 3.75 in. Find the surface area, in inches, of the cylinder. Round your answer to thousandths place.

$$SA = 2B + Ch$$

$$SA = 2\pi(5)^2 + 2\pi(5)(3.75)$$

$$SA = 50\pi + 37.5\pi$$

$$SA = 274.889 in^2$$



*How is finding lateral area different from finding surface area? (Lateral area does not involve the bases.)*



#### *Common Misconception*

*Students may not round to the correct place value and may require additional help and scaffolding with rounding. Provide this to students as necessary. This may be an issue with students' final answers.*

## 9.8.8 Wrap-Up: Error Analysis

**Component Overview:** Students complete an error analysis of a question involving the surface area of a square pyramid.



### 9.8.8 Wrap-Up: Error Analysis

1. Cody and Mila are finding the surface area of a square pyramid whose base area is 144 square inches and slant height is 16 inches. Their work is shown in the table.

| Cody's Work                      | Mila's Work                         |
|----------------------------------|-------------------------------------|
| $SA = B + \frac{1}{2}Ph_s$       | $SA = B + \frac{1}{2}Ph_s$          |
| $SA = 144 + \frac{1}{2}(48)(16)$ | $SA = (144) + \frac{1}{2}(144)(16)$ |
| $SA = 528$ square inches         | $SA = 1296$ square inches           |

- a. Whose work is correct?  
*Cody is correct.*
- b. Write a note to the student whose work is incorrect describing the mistake the student made.  
*Mila, your process is correct; you used the correct formula. However, in your first step, you substituted, 144, the measure of the area of the base of the pyramid instead of, 48, the perimeter of the base.*



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



For an extension, change a part of Cody's work that would make his work incorrect.